

Field Gateway Connectivity and Telemetry Degradation Investigation

Investigation of degraded behavior observed on field gateway GW-204 at NORTH-RIDGE-17. The session shows two elevated periods involving cellular signal loss, repeated reconnect attempts, Modbus timeouts, telemetry upload failures, queue growth, and delayed backlog processing. Findings focus on the progression from initial connectivity degradation through downstream buffering and worker-stall effects, with the known-good gateway session used as baseline context for comparison.

Generated 2026-10-01T19:30:19.166Z

SOURCES 2	COMPARISONS 1	CHRONOLOGY ENTRIES 24	TECHNICAL APPENDIX Included
--------------	------------------	--------------------------	--------------------------------

Local workstation source paths are intentionally omitted from the shareable report snapshot.

Sources

Source	Importer	Processed	Imported	Skipped	Truncated
field-gateway-known-good.log	key-value	203	203	0	No
field-gateway-degraded.log	key-value	222	222	0	No

Cross-source chronology reflects timestamps as imported by TraceScope. Clock synchronization or clock skew between independent sources is not inferred.

Time Coverage

Source	Timestamped records	First timestamp	Last timestamp
field-gateway-known-good.log	203	2026-08-20T16:00:00.000Z	2026-08-20T16:40:00.000Z
field-gateway-degraded.log	222	2026-08-20T16:00:00.000Z	2026-08-20T16:40:00.000Z

Evidence Chronology

Only timestamped report evidence appears here. Untimestamped evidence remains with its source section.

Show 24 chronology entries

Timestamp	2026-08-20T16:10:31.000Z
Source	field-gateway-known-good.log
Severity	Warning
Message	Single Modbus retry completed successfully
Context	Finding: Resolved, Note

Timestamp	2026-08-20T16:11:03.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Cellular signal dropped below operating margin
Context	Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:11:21.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Cellular uplink reconnect attempt started
Context	Burst evidence

Timestamp	2026-08-20T16:11:39.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Flow meter register read timed out
Context	Burst evidence

Timestamp	2026-08-20T16:11:57.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Telemetry upload retry scheduled
Context	Burst evidence

Timestamp	2026-08-20T16:12:15.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Cellular uplink disconnected
Context	Bookmarked, Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:12:33.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Local telemetry buffer depth exceeded target
Context	Burst evidence

Timestamp	2026-08-20T16:12:51.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Second flow meter read timed out
Context	Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:13:09.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Upstream synchronization delayed by link instability
Context	Burst evidence

Timestamp	2026-08-20T16:13:27.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Gateway began another reconnect attempt
Context	Burst evidence

Timestamp	2026-08-20T16:13:45.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Reconnect attempt failed and link dropped again
Context	Bookmarked, Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:14:03.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Buffered flow telemetry queued for retry
Context	Burst evidence

Timestamp	2026-08-20T16:25:47.000Z
Source	field-gateway-known-good.log
Severity	Warning
Message	Transient signal dip recovered without disconnect
Context	Finding: Resolved, Note

Timestamp	2026-08-20T16:26:36.000Z
Source	field-gateway-known-good.log
Severity	Info
Message	Telemetry batch uploaded successfully
Context	Finding: Dismissed, Note

Timestamp	2026-08-20T16:28:03.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Residual telemetry backlog began growing again
Context	Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:28:21.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Buffered telemetry upload retry scheduled
Context	Burst evidence

Timestamp	2026-08-20T16:28:39.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Buffered telemetry upload timed out
Context	Bookmarked, Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:28:57.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Local buffer write latency exceeded target
Context	Burst evidence

Timestamp	2026-08-20T16:29:15.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Telemetry worker exceeded watchdog timing target
Context	Burst evidence

Timestamp	2026-08-20T16:29:33.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Second buffered upload timed out
Context	Burst evidence

Timestamp	2026-08-20T16:29:51.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Telemetry buffer continued growing during retries
Context	Burst evidence

Timestamp	2026-08-20T16:30:09.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Flow telemetry batch queued behind backlog
Context	Burst evidence

Timestamp	2026-08-20T16:30:27.000Z
Source	field-gateway-degraded.log
Severity	Error
Message	Telemetry worker stalled during backlog drain
Context	Bookmarked, Finding: Open, Note, Burst evidence

Timestamp	2026-08-20T16:30:45.000Z
Source	field-gateway-degraded.log
Severity	Warning
Message	Buffer persistence remained slower than target
Context	Finding: Resolved, Note, Burst evidence

field-gateway-known-good → field-gateway-degraded

BASELINE
field-gateway-known-good.log
Scope: Complete imported session

→

COMPARISON
field-gateway-degraded.log
Scope: Complete imported session

BASELINE RECORDS
203

COMPARISON RECORDS
222

RECORD DELTA
+19

Captured Timing

	Baseline	Comparison
Timestamped records	203	222
First timestamp	2026-08-20T16:00:00.000Z	2026-08-20T16:00:00.000Z
Last timestamp	2026-08-20T16:40:00.000Z	2026-08-20T16:40:00.000Z
Duration	40.0 min	40.0 min
Observed record rate	5.08 records/min	5.55 records/min

Severity Changes

Severity	Baseline	Comparison	Delta
Warning	2	14	+12
Error	0	7	+7

Event-Code Changes

Value	Baseline	Comparison	Delta
APPEARED			
UPLOAD_RETRY	0	4	+4
BUFFER_GROWING	0	3	+3
LINK_DROP	0	2	+2
LINK_RETRY	0	2	+2
MODBUS_TIMEOUT	0	2	+2
PROCESS_STALL	0	2	+2
UPLOAD_TIMEOUT	0	2	+2
WRITE_SLOW	0	2	+2
SIGNAL_LOW	0	1	+1
SYNC_DELAYED	0	1	+1
DISAPPEARED			
TRANSIENT_RECOVERY	2	0	-2

Elevated Subsystem Changes

Value	Baseline	Comparison	Delta
Telemetry	0	6	+6
Storage	0	5	+5
Network	2	5	+3
Modbus	0	2	+2
Watchdog	0	2	+2
Sync	0	1	+1

Elevated Entity Changes

Value	Baseline	Comparison	Delta
PLC-03	0	13	+13
FLOW-12	0	4	+4
RTU-07	0	4	+4
PUMP-01	1	0	-1
PUMP-02	1	0	-1

Custom-Field Changes — Categorical Fields

Field	Value	Baseline	Comparison	Change
Firmware	3.14.8-rc1	0	222	Appeared
Firmware	3.14.7	203	0	Disappeared

Custom-Field Changes — Numeric Fields

Field	Baseline	Comparison
Disk Usage (%)	min 41 · median 43 · max 46 · n=203	min 41 · median 44 · max 46 · n=222
Latency (ms)	min 18 · median 34 · max 86 · n=203	min 18 · median 34 · max 850 · n=222
Queue Depth	min 2 · median 4 · max 6 · n=203	min 2 · median 4 · max 46 · n=222
Reconnect Count	min 0 · median 0 · max 0 · n=203	min 0 · median 0 · max 3 · n=222
RSSI (dBm)	min -74 · median -65 · max -63 · n=203	min -92 · median -65 · max -63 · n=222

Burst Comparison

Shared burst-detection settings

Window	5.0 min
Elevated-event threshold	5
Error/Critical threshold	3
Merge gap	1.0 min

Session	Baseline
Bursts	0
Elevated records	0
Peak elevated	0
Longest burst	0 ms
Dominant subsystem	Unavailable
Dominant event code	Unavailable
Dominant entity	Unavailable

Session	Comparison
Bursts	2
Elevated records	21
Peak elevated	11
Longest burst	3.0 min
Dominant subsystem	Telemetry (6)
Dominant event code	UPLOAD_RETRY (4)
Dominant entity	PLC-03 (13)

This comparison is the immutable analysis captured by the TraceScope comparison document. It was not recalculated from current session state during report generation.

field-gateway-known-good.log

field-gateway-known-good.log · key-value

SOURCE RECORDS

203

VISIBLE RECORDS

203

ANALYSIS RECORDS

203

EVIDENCE RECORDS

3

Investigation Scope

Total imported population	203
Visible population	203
Visible timestamped records	203
Visible first timestamp	2026-08-20T16:00:00.000Z
Visible last timestamp	2026-08-20T16:40:00.000Z
Analysis population	203
Analysis timestamped records	203
Analysis first timestamp	2026-08-20T16:00:00.000Z
Analysis last timestamp	2026-08-20T16:40:00.000Z

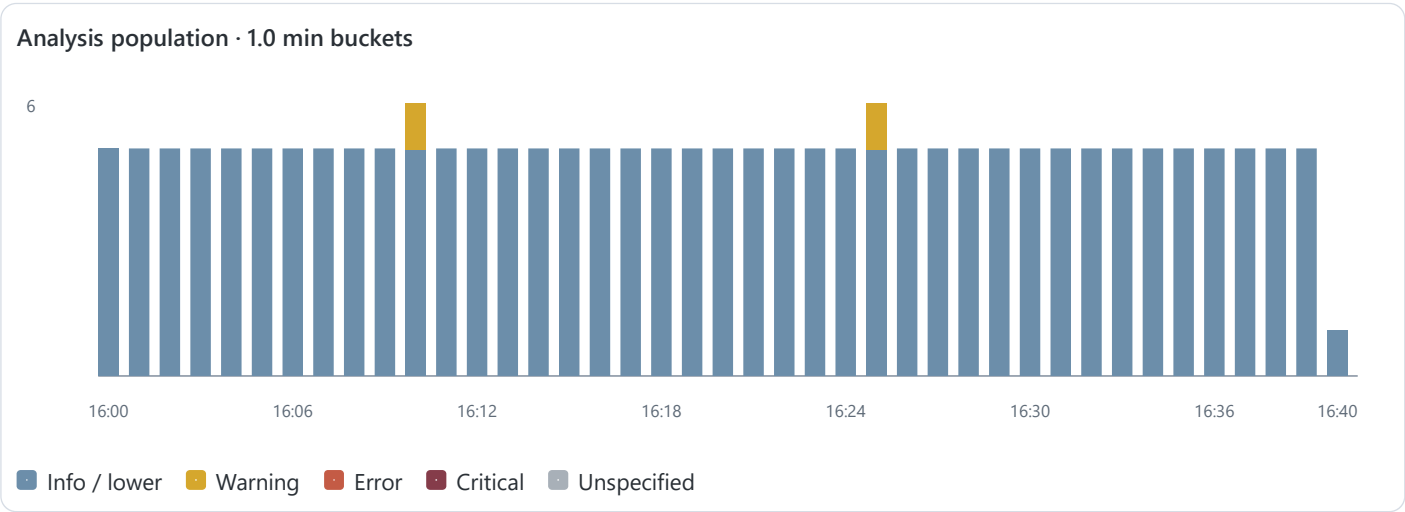
Analysis populations intentionally ignore annotation-only bookmark and finding-status filters. The visible population preserves the complete current table filter state.

Severity Summary

Severity	Visible	Analysis
Trace	0	0
Debug	67	67
Info	134	134
Warning	2	2
Error	0	0
Critical	0	0
Unspecified	0	0
Total	203	203

Event Activity Timeline

The timeline represents the deterministic analysis population captured at export time.



Grouped Warning/Error Analysis — Subsystem Issue Groups (1)

Subsystem	Warnings	Errors / Critical	Total elevated
Network	2	0	2

Deterministic Analytics — Event Codes (7)

These frequency summaries describe populated canonical values in the captured analysis population.

Value	Count
LINK_HEALTH	34
MODBUS_READ_OK	34
UPLOAD_OK	34
BUFFER_STABLE	33
SYNC_OK	33
WORKER_HEALTH	33
TRANSIENT_RECOVERY	2

Deterministic Analytics — Entities (6)

Value	Count
PUMP-01	35
PUMP-02	35
RTU-07	34
FLOW-12	33
PLC-03	33
PRESS-04	33

Deterministic Analytics — Subsystems (6)

Value	Count
Network	36
Modbus	34
Telemetry	34
Storage	33
Sync	33
Watchdog	33

Timestamp Cadence

TIMESTAMPED RECORDS 203	POSITIVE GAPS 202	ZERO GAPS 0	MEDIAN GAP 12.0 s
---	---	-----------------------------------	---

Cadence statistics

Minimum positive gap	1.0 s
Median positive gap	12.0 s
Mean positive gap	11.9 s
90th percentile gap	12.0 s
Maximum positive gap	12.0 s
Recommended burst window	5.0 min
Recommended merge gap	1.0 min
Recommendation basis	Observed timestamp cadence

Burst Analysis

DETECTED BURSTS

0

TIMING MODE

Auto

EFFECTIVE WINDOW

5.0 min

EFFECTIVE MERGE GAP

1.0 min

Burst detection settings

Configured

Window	30.0 s
Elevated-event threshold	5
Error/Critical threshold	3
Merge gap	5.0 s

Effective

Window	5.0 min
Elevated-event threshold	5
Error/Critical threshold	3
Merge gap	1.0 min

Burst analysis was available, but no bursts met the captured detection thresholds.

Findings and Investigator Annotations

Finding	Resolved
Severity	Warning
Message	Single Modbus retry completed successfully
Note	Single transient retry completed successfully without sustained connectivity or queue impact.
Bookmark	No
Timestamp	2026-08-20T16:10:31.000Z
Event #	54

Finding	Resolved
Severity	Warning
Message	Transient signal dip recovered without disconnect
Note	Transient signal dip recovered without disconnect or continued degradation.
Bookmark	No
Timestamp	2026-08-20T16:25:47.000Z
Event #	131

Finding	Dismissed
Severity	Info
Message	Telemetry batch uploaded successfully
Note	Reviewed as baseline context; normal service behavior resumed immediately after the transient condition.
Bookmark	No
Timestamp	2026-08-20T16:26:36.000Z
Event #	136

Supporting Evidence

Includes investigator-state records and additional captured burst evidence where applicable.

Warning · Single Modbus retry completed successfully

Finding: Resolved	Note
Event #	54
Timestamp	2026-08-20T16:10:31.000Z
Severity	Warning
Subsystem	Network
Event code	TRANSIENT_RECOVERY
Entity ID	PUMP-02
Message	Single Modbus retry completed successfully
Source	field-gateway-known-good.log
Source record	54
Source generation	0
Record ID	ec1fd135bb3ddc5bba80f0b0e64a4b1a59a27b38985ebedc8f90338b6f99878a

Analyst Note

Single transient retry completed successfully without sustained connectivity or queue impact.

Custom Attributes

Disk Usage (%)	43
Firmware	3.14.7
Gateway	GW-204
Interface	cell0
Latency (ms)	86
Queue Depth	4
Reconnect Count	0
RSSI (dBm)	-74
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:10:31.000Z level=WARN subsystem=Network eventCode=TRANSIENT_RECOVERY
entityId=PUMP-02 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.7 interface=cell0 latencyMs=86
queueDepth=4 reconnectCount=0 rssiDbm=-74 diskUsagePct=43 message="Single Modbus retry completed
successfully"
```

Warning · Transient signal dip recovered without disconnect

Finding: Resolved Note

Event #	131
Timestamp	2026-08-20T16:25:47.000Z
Severity	Warning
Subsystem	Network
Event code	TRANSIENT_RECOVERY
Entity ID	PUMP-01
Message	Transient signal dip recovered without disconnect
Source	field-gateway-known-good.log
Source record	131
Source generation	0
Record ID	6b3248d457ca7fb780e2f5f613edfca11b73da2ea02e7690cc4208148e9aecff

Analyst Note

Transient signal dip recovered without disconnect or continued degradation.

Custom Attributes

Disk Usage (%)	43
Firmware	3.14.7
Gateway	GW-204
Interface	cell0
Latency (ms)	86
Queue Depth	4
Reconnect Count	0
RSSI (dBm)	-74
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:25:47.000Z level=WARN subsystem=Network eventCode=TRANSIENT_RECOVERY
entityId=PUMP-01 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.7 interface=cell0 latencyMs=86
queueDepth=4 reconnectCount=0 rssiDbm=-74 diskUsagePct=43 message="Transient signal dip recovered without
disconnect"
```

Info · Telemetry batch uploaded successfully

Finding: Dismissed Note

Event #	136
Timestamp	2026-08-20T16:26:36.000Z
Severity	Info
Subsystem	Telemetry
Event code	UPLOAD_OK
Entity ID	PUMP-02
Message	Telemetry batch uploaded successfully
Source	field-gateway-known-good.log
Source record	136
Source generation	0
Record ID	3679c55568277bc29e41672d49ac75ec37cfabb73d69a1888d05536cf0f1d058

Analyst Note

Reviewed as baseline context; normal service behavior resumed immediately after the transient condition.

Custom Attributes

Disk Usage (%)	42
Firmware	3.14.7
Gateway	GW-204
Interface	cell0
Latency (ms)	46
Queue Depth	5
Reconnect Count	0
RSSI (dBm)	-64
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:26:36.000Z level=INFO subsystem=Telemetry eventCode=UPLOAD_OK entityId=PUMP-02
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.7 interface=cell0 latencyMs=46 queueDepth=5
reconnectCount=0 rssiDbm=-64 diskUsagePct=42 message="Telemetry batch uploaded successfully"
```

Import and Data-Quality Context

Source name	field-gateway-known-good.log
Source size	57.9 KiB
Source last modified	2026-08-23T15:27:44.261Z
Imported at	2026-10-01T19:15:19.057Z
Importer	key-value
Import profile	Field Gateway Support Session
Processed records	203
Imported records	203
Skipped records	0
Source truncated	No

Import Diagnostics (0)

No import diagnostics were captured.

Complete Import Profile

Schema version	1
Profile name	Field Gateway Support Session
Importer	key-value
Preserve unmapped fields	Yes

Canonical Field Mappings

Canonical field	Source path
Timestamp	timestamp
Severity	level
Subsystem	subsystem
Event code	eventCode
Entity ID	entityId
Message	message

Custom Field Mappings (9)

Name	Source path
Site	site
Gateway	gateway
Firmware	firmware
Interface	interface
Latency (ms)	latencyMs
Queue Depth	queueDepth
Reconnect Count	reconnectCount
RSSI (dBm)	rssiDbm
Disk Usage (%)	diskUsagePct

Severity Aliases (0)

No severity aliases were configured.

Timestamp Rules (1)

Priority	Type	Format
1	ISO 8601	Default

Captured investigation filters

No investigation filters were active when this report snapshot was captured.

Source capabilities

Timestamps	Available
Severity	Available
Subsystem	Available
Event code	Available
Entity	Available
Custom fields	Available

--

field-gateway-degraded.log

field-gateway-degraded.log · key-value

SOURCE RECORDS

222

VISIBLE RECORDS

222

ANALYSIS RECORDS

222

EVIDENCE RECORDS

21

Investigation Scope

Total imported population	222
Visible population	222
Visible timestamped records	222
Visible first timestamp	2026-08-20T16:00:00.000Z
Visible last timestamp	2026-08-20T16:40:00.000Z
Analysis population	222
Analysis timestamped records	222
Analysis first timestamp	2026-08-20T16:00:00.000Z
Analysis last timestamp	2026-08-20T16:40:00.000Z

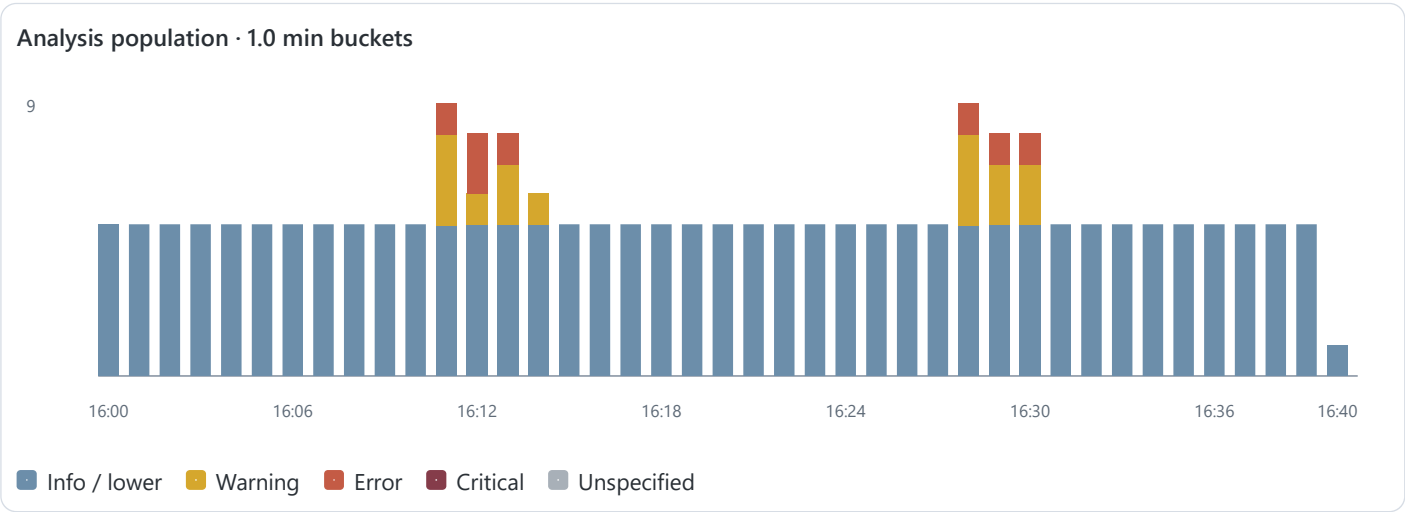
Analysis populations intentionally ignore annotation-only bookmark and finding-status filters. The visible population preserves the complete current table filter state.

Severity Summary

Severity	Visible	Analysis
Trace	0	0
Debug	67	67
Info	134	134
Warning	14	14
Error	7	7
Critical	0	0
Unspecified	0	0
Total	222	222

Event Activity Timeline

The timeline represents the deterministic analysis population captured at export time.



Grouped Warning/Error Analysis — Subsystem Issue Groups (6)

Subsystem	Warnings	Errors / Critical	Total elevated
Telemetry	4	2	6
Network	3	2	5
Storage	5	0	5
Modbus	0	2	2
Watchdog	1	1	2
Sync	1	0	1

Deterministic Analytics — Event Codes (16)

These frequency summaries describe populated canonical values in the captured analysis population.

Value	Count
LINK_HEALTH	34
MODBUS_READ_OK	34
UPLOAD_OK	34
BUFFER_STABLE	33
SYNC_OK	33
WORKER_HEALTH	33
UPLOAD_RETRY	4
BUFFER_GROWING	3
LINK_DROP	2
LINK_RETRY	2
MODBUS_TIMEOUT	2
PROCESS_STALL	2
UPLOAD_TIMEOUT	2
WRITE_SLOW	2
SIGNAL_LOW	1
SYNC_DELAYED	1

Deterministic Analytics — Entities (6)

Value	Count
PLC-03	46
RTU-07	38
FLOW-12	37
PUMP-01	34
PUMP-02	34
PRESS-04	33

Deterministic Analytics — Subsystems (6)

Value	Count
Telemetry	40
Network	39
Storage	38
Modbus	36
Watchdog	35
Sync	34

Timestamp Cadence

TIMESTAMPED RECORDS 222	POSITIVE GAPS 221	ZERO GAPS 0	MEDIAN GAP 12.0 s
---	---	-----------------------------------	---

Cadence statistics

Minimum positive gap	3.0 s
Median positive gap	12.0 s
Mean positive gap	10.9 s
90th percentile gap	12.0 s
Maximum positive gap	12.0 s
Recommended burst window	5.0 min
Recommended merge gap	1.0 min
Recommendation basis	Observed timestamp cadence

Burst Analysis

DETECTED BURSTS

2

TIMING MODE

Auto

EFFECTIVE WINDOW

5.0 min

EFFECTIVE MERGE GAP

1.0 min

Burst detection settings

Configured

Window	30.0 s
Elevated-event threshold	5
Error/Critical threshold	3
Merge gap	5.0 s

Effective

Window	5.0 min
Elevated-event threshold	5
Error/Critical threshold	3
Merge gap	1.0 min

Detected Bursts (2)

Start	2026-08-20T16:11:03.000Z
End	2026-08-20T16:14:03.000Z
Duration	3.0 min
Warning	7
Error	4
Critical	0
Total elevated	11
Trigger	Elevated-event threshold, Error/Critical threshold

Start	2026-08-20T16:28:03.000Z
End	2026-08-20T16:30:45.000Z
Duration	2.7 min
Warning	7
Error	3
Critical	0
Total elevated	10
Trigger	Elevated-event threshold, Error/Critical threshold

Findings and Investigator Annotations

Finding	Open
Severity	Warning
Message	Cellular signal dropped below operating margin
Note	Cellular signal fell below operating margin at the start of the first degraded period.
Bookmark	No
Timestamp	2026-08-20T16:11:03.000Z
Event #	57

Finding	Open
Severity	Error
Message	Cellular uplink disconnected
Note	Cellular uplink disconnected after repeated reconnect activity; RSSI reached -91 dBm and queue depth had risen to 15.
Bookmark	Yes
Timestamp	2026-08-20T16:12:15.000Z
Event #	67

Finding	Open
Severity	Error
Message	Second flow meter read timed out
Note	Second flow-meter timeout occurred while the network remained unstable and buffered work continued accumulating.
Bookmark	No
Timestamp	2026-08-20T16:12:51.000Z
Event #	72

Finding	Open
Severity	Error
Message	Reconnect attempt failed and link dropped again
Note	Reconnect attempt failed and the cellular link dropped again; queue depth had increased to 34.
Bookmark	Yes
Timestamp	2026-08-20T16:13:45.000Z
Event #	79

Finding	Open
Severity	Warning
Message	Residual telemetry backlog began growing again
Note	Residual telemetry backlog began growing again after the earlier connectivity incident appeared to recover.
Bookmark	No
Timestamp	2026-08-20T16:28:03.000Z
Event #	153

Finding	Open
Severity	Error
Message	Buffered telemetry upload timed out
Note	Buffered telemetry upload timed out as the secondary backlog period intensified.
Bookmark	Yes
Timestamp	2026-08-20T16:28:39.000Z
Event #	158

Finding	Open
Severity	Error
Message	Telemetry worker stalled during backlog drain
Note	Telemetry worker stalled during backlog drain with queue depth reaching 46.
Bookmark	Yes
Timestamp	2026-08-20T16:30:27.000Z
Event #	173

Finding	Resolved
Severity	Warning
Message	Buffer persistence remained slower than target
Note	Buffer persistence remained slower than target during the backlog-drain period.
Bookmark	No
Timestamp	2026-08-20T16:30:45.000Z
Event #	175

Supporting Evidence

Includes investigator-state records and additional captured burst evidence where applicable.

Warning · Cellular signal dropped below operating margin

Finding: Open	Note	Burst evidence
Event #	57	
Timestamp	2026-08-20T16:11:03.000Z	
Severity	Warning	
Subsystem	Network	
Event code	SIGNAL_LOW	
Entity ID	PLC-03	
Message	Cellular signal dropped below operating margin	
Source	field-gateway-degraded.log	
Source record	57	
Source generation	0	
Record ID	4c9dd7a8771b7d36f59e099df6a216703d0cfb166b6a45d8d22cd0a899f55cbe	

Analyst Note

Cellular signal fell below operating margin at the start of the first degraded period.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	6
Reconnect Count	1
RSSI (dBm)	-84
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:11:03.000Z level=WARN subsystem=Network eventCode=SIGNAL_LOW entityId=PLC-03
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=6
reconnectCount=1 rssiDbm=-84 diskUsagePct=46 message="Cellular signal dropped below operating margin"
```

Warning · Cellular uplink reconnect attempt started

Burst evidence	
Event #	59
Timestamp	2026-08-20T16:11:21.000Z
Severity	Warning
Subsystem	Network
Event code	LINK_RETRY
Entity ID	PLC-03
Message	Cellular uplink reconnect attempt started
Source	field-gateway-degraded.log
Source record	59
Source generation	0
Record ID	5d54a0ae39ed6c7d59e186a0b29f9095eee418cc4123b4461bce94e885c7af9e
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	8
Reconnect Count	1
RSSI (dBm)	-86
Site	NORTH-RIDGE-17
Raw Source	
<pre>timestamp=2026-08-20T16:11:21.000Z level=WARN subsystem=Network eventCode=LINK_RETRY entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=8 reconnectCount=1 rssiDbm=-86 diskUsagePct=46 message="Cellular uplink reconnect attempt started"</pre>	

Error · Flow meter register read timed out

Burst evidence	
Event #	62
Timestamp	2026-08-20T16:11:39.000Z
Severity	Error
Subsystem	Modbus
Event code	MODBUS_TIMEOUT
Entity ID	FLOW-12
Message	Flow meter register read timed out
Source	field-gateway-degraded.log
Source record	62
Source generation	0
Record ID	babf7b4996a9035c9a5fcb61cc34a5ab10f361675cfdabf8a426ec7ad365229c
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	850
Queue Depth	9
Reconnect Count	1
RSSI (dBm)	-82
Site	NORTH-RIDGE-17
Raw Source	
timestamp=2026-08-20T16:11:39.000Z level=ERROR subsystem=Modbus eventCode=MODBUS_TIMEOUT entityId=FLOW-12 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=850 queueDepth=9 reconnectCount=1 rssiDbm=-82 diskUsagePct=46 message="Flow meter register read timed out"	

Warning · Telemetry upload retry scheduled

Burst evidence	
Event #	64
Timestamp	2026-08-20T16:11:57.000Z
Severity	Warning
Subsystem	Telemetry
Event code	UPLOAD_RETRY
Entity ID	PLC-03
Message	Telemetry upload retry scheduled
Source	field-gateway-degraded.log
Source record	64
Source generation	0
Record ID	bd209e356e9a5ddc01412e84f2482439c2dd78136abe6c21d6978f56558ae0b2
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	12
Reconnect Count	2
RSSI (dBm)	-85
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:11:57.000Z level=WARN subsystem=Telemetry eventCode=UPLOAD_RETRY entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=12 reconnectCount=2 rssiDbm=-85 diskUsagePct=46 message="Telemetry upload retry scheduled"

Error · Cellular uplink disconnected

Bookmarked	Finding: Open	Note	Burst evidence
Event #			67
Timestamp			2026-08-20T16:12:15.000Z
Severity			Error
Subsystem			Network
Event code			LINK_DROP
Entity ID			PLC-03
Message			Cellular uplink disconnected
Source			field-gateway-degraded.log
Source record			67
Source generation			0
Record ID			c8752bffcf19a0cb0f2bc0ee4f5401a74329eb3eef7c85489cb6c8b7e6aac9f3

Analyst Note

Cellular uplink disconnected after repeated reconnect activity; RSSI reached -91 dBm and queue depth had risen to 15.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	850
Queue Depth	15
Reconnect Count	2
RSSI (dBm)	-91
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:12:15.000Z level=ERROR subsystem=Network eventCode=LINK_DROP entityId=PLC-03
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=850 queueDepth=15
reconnectCount=2 rssiDbm=-91 diskUsagePct=46 message="Cellular uplink disconnected"
```

Warning · Local telemetry buffer depth exceeded target

Burst evidence	
Event #	69
Timestamp	2026-08-20T16:12:33.000Z
Severity	Warning
Subsystem	Storage
Event code	BUFFER_GROWING
Entity ID	PLC-03
Message	Local telemetry buffer depth exceeded target
Source	field-gateway-degraded.log
Source record	69
Source generation	0
Record ID	f89aa2c25dc7edd0266a52e3f0da2a32d3d7b891b064efc9c740678b7ace12e9
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	21
Reconnect Count	2
RSSI (dBm)	-89
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:12:33.000Z level=WARN subsystem=Storage eventCode=BUFFER_GROWING entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=21 reconnectCount=2 rssiDbm=-89 diskUsagePct=46 message="Local telemetry buffer depth exceeded target"

Error · Second flow meter read timed out

Finding: Open Note Burst evidence

Event #	72
Timestamp	2026-08-20T16:12:51.000Z
Severity	Error
Subsystem	Modbus
Event code	MODBUS_TIMEOUT
Entity ID	FLOW-12
Message	Second flow meter read timed out
Source	field-gateway-degraded.log
Source record	72
Source generation	0
Record ID	41bd20bac47579634384d744c58610152ab571cb3ad2198cd5da835c6d621cdd

Analyst Note

Second flow-meter timeout occurred while the network remained unstable and buffered work continued accumulating.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	850
Queue Depth	24
Reconnect Count	2
RSSI (dBm)	-86
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:12:51.000Z level=ERROR subsystem=Modbus eventCode=MODBUS_TIMEOUT entityId=FLOW-12
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=850 queueDepth=24
reconnectCount=2 rssiDbm=-86 diskUsagePct=46 message="Second flow meter read timed out"
```

Warning · Upstream synchronization delayed by link instability

Burst evidence

Event #	74
Timestamp	2026-08-20T16:13:09.000Z
Severity	Warning
Subsystem	Sync
Event code	SYNC_DELAYED
Entity ID	RTU-07
Message	Upstream synchronization delayed by link instability
Source	field-gateway-degraded.log
Source record	74
Source generation	0
Record ID	e5900adb93cf63a62776955a144bb890ad6ca57a211541f5c344490e8010f5ab
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	27
Reconnect Count	2
RSSI (dBm)	-84
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:13:09.000Z level=WARN subsystem=Sync eventCode=SYNC_DELAYED entityId=RTU-07
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=27
reconnectCount=2 rssiDbm=-84 diskUsagePct=46 message="Upstream synchronization delayed by link
instability"
```

Warning · Gateway began another reconnect attempt

Burst evidence	
Event #	77
Timestamp	2026-08-20T16:13:27.000Z
Severity	Warning
Subsystem	Network
Event code	LINK_RETRY
Entity ID	PLC-03
Message	Gateway began another reconnect attempt
Source	field-gateway-degraded.log
Source record	77
Source generation	0
Record ID	d3e85ef43972cfabcce9b0ec0bcba0fda9cc07b9fe37d06b797dae2133bc58be
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	30
Reconnect Count	3
RSSI (dBm)	-88
Site	NORTH-RIDGE-17
Raw Source	
timestamp=2026-08-20T16:13:27.000Z level=WARN subsystem=Network eventCode=LINK_RETRY entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=30 reconnectCount=3 rssiDbm=-88 diskUsagePct=46 message="Gateway began another reconnect attempt"	

Error · Reconnect attempt failed and link dropped again

Bookmarked	Finding: Open	Note	Burst evidence
Event #			79
Timestamp			2026-08-20T16:13:45.000Z
Severity			Error
Subsystem			Network
Event code			LINK_DROP
Entity ID			PLC-03
Message			Reconnect attempt failed and link dropped again
Source			field-gateway-degraded.log
Source record			79
Source generation			0
Record ID			0984f388b33f174c135d8a6536e9efae7c09b107de886c05c386c1e4c4ff151f

Analyst Note

Reconnect attempt failed and the cellular link dropped again; queue depth had increased to 34.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	850
Queue Depth	34
Reconnect Count	3
RSSI (dBm)	-92
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:13:45.000Z level=ERROR subsystem=Network eventCode=LINK_DROP entityId=PLC-03
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=850 queueDepth=34
reconnectCount=3 rssiDbm=-92 diskUsagePct=46 message="Reconnect attempt failed and link dropped again"
```

Warning · Buffered flow telemetry queued for retry

Burst evidence	
Event #	82
Timestamp	2026-08-20T16:14:03.000Z
Severity	Warning
Subsystem	Telemetry
Event code	UPLOAD_RETRY
Entity ID	FLOW-12
Message	Buffered flow telemetry queued for retry
Source	field-gateway-degraded.log
Source record	82
Source generation	0
Record ID	59a394bef9bb5bef3efd6d086302d38682d79d342e2cdfb5b45961f54cf97955
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	37
Reconnect Count	3
RSSI (dBm)	-85
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:14:03.000Z level=WARN subsystem=Telemetry eventCode=UPLOAD_RETRY entityId=FLOW-12 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=37 reconnectCount=3 rssiDbm=-85 diskUsagePct=46 message="Buffered flow telemetry queued for retry"

Warning · Residual telemetry backlog began growing again

Finding: Open Note Burst evidence

Event #	153
Timestamp	2026-08-20T16:28:03.000Z
Severity	Warning
Subsystem	Storage
Event code	BUFFER_GROWING
Entity ID	RTU-07
Message	Residual telemetry backlog began growing again
Source	field-gateway-degraded.log
Source record	153
Source generation	0
Record ID	394c1af394c2026d76b1cc3dd2304b191df95369a94ebe443e7cbc36810c4162

Analyst Note

Residual telemetry backlog began growing again after the earlier connectivity incident appeared to recover.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	18
Reconnect Count	3
RSSI (dBm)	-70
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:28:03.000Z level=WARN subsystem=Storage eventCode=BUFFER_GROWING entityId=RTU-07
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=18
reconnectCount=3 rssiDbm=-70 diskUsagePct=46 message="Residual telemetry backlog began growing again"
```

Warning · Buffered telemetry upload retry scheduled

Burst evidence	
Event #	155
Timestamp	2026-08-20T16:28:21.000Z
Severity	Warning
Subsystem	Telemetry
Event code	UPLOAD_RETRY
Entity ID	RTU-07
Message	Buffered telemetry upload retry scheduled
Source	field-gateway-degraded.log
Source record	155
Source generation	0
Record ID	1bd2728d5b3eefec60691a6398d42de9301905409dedb50b62e2062c9229a77a
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	22
Reconnect Count	3
RSSI (dBm)	-72
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:28:21.000Z level=WARN subsystem=Telemetry eventCode=UPLOAD_RETRY entityId=RTU-07 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=22 reconnectCount=3 rssiDbm=-72 diskUsagePct=46 message="Buffered telemetry upload retry scheduled"

Error · Buffered telemetry upload timed out

Bookmarked	Finding: Open	Note	Burst evidence
Event #			158
Timestamp			2026-08-20T16:28:39.000Z
Severity			Error
Subsystem			Telemetry
Event code			UPLOAD_TIMEOUT
Entity ID			PLC-03
Message			Buffered telemetry upload timed out
Source			field-gateway-degraded.log
Source record			158
Source generation			0
Record ID			0da4926a710fe3020deecae9cc81a6929e567db78b2047fad72895700b3c19c0

Analyst Note

Buffered telemetry upload timed out as the secondary backlog period intensified.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	850
Queue Depth	28
Reconnect Count	3
RSSI (dBm)	-74
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:28:39.000Z level=ERROR subsystem=Telemetry eventCode=UPLOAD_TIMEOUT
entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=850
queueDepth=28 reconnectCount=3 rssiDbm=-74 diskUsagePct=46 message="Buffered telemetry upload timed out"
```

Warning · Local buffer write latency exceeded target

Burst evidence	
Event #	160
Timestamp	2026-08-20T16:28:57.000Z
Severity	Warning
Subsystem	Storage
Event code	WRITE_SLOW
Entity ID	PLC-03
Message	Local buffer write latency exceeded target
Source	field-gateway-degraded.log
Source record	160
Source generation	0
Record ID	0a3365890ef12872df0589b34ce32ae5b692ad6dc7343f88ebde1cc401030763
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	31
Reconnect Count	3
RSSI (dBm)	-72
Site	NORTH-RIDGE-17
Raw Source	
timestamp=2026-08-20T16:28:57.000Z level=WARN subsystem=Storage eventCode=WRITE_SLOW entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=31 reconnectCount=3 rssiDbm=-72 diskUsagePct=46 message="Local buffer write latency exceeded target"	

Warning · Telemetry worker exceeded watchdog timing target

Burst evidence	
Event #	163
Timestamp	2026-08-20T16:29:15.000Z
Severity	Warning
Subsystem	Watchdog
Event code	PROCESS_STALL
Entity ID	PLC-03
Message	Telemetry worker exceeded watchdog timing target
Source	field-gateway-degraded.log
Source record	163
Source generation	0
Record ID	89a675b233e19d9dc8e4aeeb60e1d54477e1810380b4bd9418e4ace114133c5c
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	34
Reconnect Count	3
RSSI (dBm)	-73
Site	NORTH-RIDGE-17
Raw Source	
timestamp=2026-08-20T16:29:15.000Z level=WARN subsystem=Watchdog eventCode=PROCESS_STALL entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=34 reconnectCount=3 rssiDbm=-73 diskUsagePct=46 message="Telemetry worker exceeded watchdog timing target"	

Error · Second buffered upload timed out

Burst evidence	
Event #	165
Timestamp	2026-08-20T16:29:33.000Z
Severity	Error
Subsystem	Telemetry
Event code	UPLOAD_TIMEOUT
Entity ID	RTU-07
Message	Second buffered upload timed out
Source	field-gateway-degraded.log
Source record	165
Source generation	0
Record ID	e8f06365d3b72a58605b520a88a06b953b70ee70f716f116b7f25b7212dd9fa7
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	850
Queue Depth	37
Reconnect Count	3
RSSI (dBm)	-75
Site	NORTH-RIDGE-17
Raw Source	
timestamp=2026-08-20T16:29:33.000Z level=ERROR subsystem=Telemetry eventCode=UPLOAD_TIMEOUT entityId=RTU-07 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=850 queueDepth=37 reconnectCount=3 rssiDbm=-75 diskUsagePct=46 message="Second buffered upload timed out"	

Warning · Telemetry buffer continued growing during retries

Burst evidence	
Event #	168
Timestamp	2026-08-20T16:29:51.000Z
Severity	Warning
Subsystem	Storage
Event code	BUFFER_GROWING
Entity ID	PLC-03
Message	Telemetry buffer continued growing during retries
Source	field-gateway-degraded.log
Source record	168
Source generation	0
Record ID	8258c1368fb2683d9d6fd77253fbefec7a2bf29b85c8df5b300cc6c238a86ca0
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	40
Reconnect Count	3
RSSI (dBm)	-73
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:29:51.000Z level=WARN subsystem=Storage eventCode=BUFFER_GROWING entityId=PLC-03 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=40 reconnectCount=3 rssiDbm=-73 diskUsagePct=46 message="Telemetry buffer continued growing during retries"

Warning · Flow telemetry batch queued behind backlog

Burst evidence	
Event #	170
Timestamp	2026-08-20T16:30:09.000Z
Severity	Warning
Subsystem	Telemetry
Event code	UPLOAD_RETRY
Entity ID	FLOW-12
Message	Flow telemetry batch queued behind backlog
Source	field-gateway-degraded.log
Source record	170
Source generation	0
Record ID	e8332cb4f8c10031f78af1c80a6135b932837e4922c55e053eed575d432fbfc4
Custom Attributes	
Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	cell0
Latency (ms)	180
Queue Depth	43
Reconnect Count	3
RSSI (dBm)	-72
Site	NORTH-RIDGE-17
Raw Source	

timestamp=2026-08-20T16:30:09.000Z level=WARN subsystem=Telemetry eventCode=UPLOAD_RETRY entityId=FLOW-12 site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=cell0 latencyMs=180 queueDepth=43 reconnectCount=3 rssiDbm=-72 diskUsagePct=46 message="Flow telemetry batch queued behind backlog"

Error · Telemetry worker stalled during backlog drain

Bookmarked	Finding: Open	Note	Burst evidence
Event #			173
Timestamp			2026-08-20T16:30:27.000Z
Severity			Error
Subsystem			Watchdog
Event code			PROCESS_STALL
Entity ID			PLC-03
Message			Telemetry worker stalled during backlog drain
Source			field-gateway-degraded.log
Source record			173
Source generation			0
Record ID			7db740ceb581eb191d844dbd0e39ff448230ef2e93135323391627291b5c896c

Analyst Note

Telemetry worker stalled during backlog drain with queue depth reaching 46.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	850
Queue Depth	46
Reconnect Count	3
RSSI (dBm)	-74
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:30:27.000Z level=ERROR subsystem=Watchdog eventCode=PROCESS_STALL entityId=PLC-03
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=850 queueDepth=46
reconnectCount=3 rssiDbm=-74 diskUsagePct=46 message="Telemetry worker stalled during backlog drain"
```

Warning · Buffer persistence remained slower than target

Finding: Resolved	Note	Burst evidence
Event #	175	
Timestamp	2026-08-20T16:30:45.000Z	
Severity	Warning	
Subsystem	Storage	
Event code	WRITE_SLOW	
Entity ID	PLC-03	
Message	Buffer persistence remained slower than target	
Source	field-gateway-degraded.log	
Source record	175	
Source generation	0	
Record ID	1c3c3c6978f7e4ad0ce9d567854452846eeee7d606f44edbc4227dd5493308b3	

Analyst Note

Buffer persistence remained slower than target during the backlog-drain period.

Custom Attributes

Disk Usage (%)	46
Firmware	3.14.8-rc1
Gateway	GW-204
Interface	rs485-1
Latency (ms)	180
Queue Depth	41
Reconnect Count	3
RSSI (dBm)	-71
Site	NORTH-RIDGE-17

Raw Source

```
timestamp=2026-08-20T16:30:45.000Z level=WARN subsystem=Storage eventCode=WRITE_SLOW entityId=PLC-03
site=NORTH-RIDGE-17 gateway=GW-204 firmware=3.14.8-rc1 interface=rs485-1 latencyMs=180 queueDepth=41
reconnectCount=3 rssiDbm=-71 diskUsagePct=46 message="Buffer persistence remained slower than target"
```

Import and Data-Quality Context

Source name	field-gateway-degraded.log
Source size	64.2 KiB
Source last modified	2026-08-23T15:27:50.576Z
Imported at	2026-10-01T19:15:39.077Z
Importer	key-value
Import profile	Field Gateway Support Session
Processed records	222
Imported records	222
Skipped records	0
Source truncated	No

Import Diagnostics (0)

No import diagnostics were captured.

Complete Import Profile

Schema version	1
Profile name	Field Gateway Support Session
Importer	key-value
Preserve unmapped fields	Yes

Canonical Field Mappings

Canonical field	Source path
Timestamp	timestamp
Severity	level
Subsystem	subsystem
Event code	eventCode
Entity ID	entityId
Message	message

Custom Field Mappings (9)

Name	Source path
Site	site
Gateway	gateway
Firmware	firmware
Interface	interface
Latency (ms)	latencyMs
Queue Depth	queueDepth
Reconnect Count	reconnectCount
RSSI (dBm)	rssidBm
Disk Usage (%)	diskUsagePct

Severity Aliases (0)

No severity aliases were configured.

Timestamp Rules (1)

Priority	Type	Format
1	ISO 8601	Default

Captured investigation filters

No investigation filters were active when this report snapshot was captured.

Source capabilities

Timestamps	Available
Severity	Available
Subsystem	Available
Event code	Available
Entity	Available
Custom fields	Available